

The six alkali metals are placed together in group one of the periodic table.

H	He																	He																															
Li	Be	B	C	N	O	F	Ne																	Ne																									
Na	Mg	Al	Si	P	S	Cl	Ar																	Ar																									
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr																	Kr															
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe																	Xe															
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn																	Rn															
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og																	Og															
																		Lr	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu																	Lu
																		Lr	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr																	Lr

Alkali Metals

The first group of the periodic table is the alkali metals. A collection of six similar elements, these metals react vigorously with water to produce hydrogen gas and alkaline solutions, and with the oxygen in air to produce compounds called oxides. Alkali metals are highly reactive and are only found in nature combined with other elements in compounds. Pure alkali metals are shiny and soft enough to be cut with a knife.

Lithium

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Li Pure lithium is so light that it floats on water. However, if left in the open, this silvery metal will tarnish in minutes. To prevent this from happening, it has to be stored in mineral oil. Discovered in 1817, lithium has a wide range of commercial uses today. For instance, it is mixed with magnesium and aluminum to form lightweight alloys used to manufacture some aircraft and trains, and it's a key element in rechargeable batteries. Lithium is also useful in medications.



Laboratory-refined piece of pure lithium

A dull oxide layer instantly forms on this pure alkali metal when it's exposed to air.

In nature

An important source of lithium is lepidolite. This lilac-gray mineral gets its name from two Greek words: *lepidos*, which means "scales," and *lithos*, meaning "stone." It is so named because it forms flaky, scale-like crystals.



Soaking it up

Minerals containing lithium dissolve easily in water. The seas and ocean are filled with vast amounts of lithium. Sea creatures, such as the common lobster, absorb it from the water.

Powerful battery

Rechargeable lithium-ion batteries are widely used in modern electronic devices, such as smartphones. Not only are they lightweight and small; these batteries also allow people to use their cell phones for a long time between charges.

